

NATIONAL INSTITUTE OF TECHNOLOGY: TIRUCHIRAPPALLI: 620 015
DEPARTMENT OF MECHANICAL ENGINEERING

– Basics of Mechanical Engineering

Roll No.

Date: 26.11.2025

Class Quiz – II

Duration: 15 Minutes

Maximum Marks: 10

- ✓ 1. An economiser is used to preheat the water before it enters the boiler by
☒ a) Outgoing flue gases
b) Steam bled from the turbine
c) Separate heating element.
d) None of the above.
- ✓ 2. Wet steam can be completely specified by
a) Pressure only
b) Temperature only
☒ c) Dryness fraction only
d) Pressure and dryness fraction
- ✓ 3. In a Rankine cycle heat addition takes place at *Constant entropy*
☒ a) Constant temperature.
☒ b) Constant pressure
c) Constant Volume
☒ d) Constant entropy
- ✓ 4. In refrigeration cycle Heat is absorbed by refrigerant in
a) Compressor
b) Condenser
☒ c) Evaporator
d) Expansion device.
- ✓ 5. The water which is at a temperature lower than the saturation temperature of its pressure then it is called
☒ a) Subcooled liquid
b) Wet vapor
c) Saturated liquid
d) Saturated vapor.
- ✓ 6. Match the following

I. Leaf spring suspension (ii)	i. Semi-dependent suspension systems
II. Telescopic suspension (iv)	ii. Dependent suspension system
III. Compliant link	iii. Independent suspension system
IV. MacPherson Strut (iii)	iv. Two wheeler front suspension

a) I-ii, II-iv, III-iii, IV-i ✕

b) I-ii, II-i, III-iv, IV-iii

☒ c) I-ii, II-iv, III-i, IV-iii

d) I-iii, II-ii, III-i, IV-iv ✕

✓ 7. Select the correct function of a clutch:

a) To provide the torque required to move the vehicle. ✕

☒ b) To allow smooth engagement of power between engine and transmission.

c) To transmits power from the gearbox to the differential. ✕

d) All of the above ✕

8. Identify the correct classification of Al_2O_3 .

a) Metal

b) Ceramic

☒ c) Polymer

d) Composite

9. Based on the Iron-Iron Carbide equilibrium diagram, above $912^{\circ}C$, pure iron exist in _____ phase.

☒ a) Ferrite

b) Pearlite

c) Austenite

d) Cementite

10. Choose the incorrect statement regarding different types of solids.

a) Metallic bonds are responsible for ductility.

b) Ionic solids generally possess high melting points. ✓

c) Covalent network solids exhibit poor electrical conductivity. ✓

☒ d) Molecular solids are good electrical conductors.